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# Side-by-Side PowerShell and Module Environments with multi-pwsh

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# Marc-André Moreau

- Chief Technology Officer at Devolutions
- Microsoft MVP (PowerShell & Windows)
- Creator of the FreeRDP open source project
- I love PowerShell, .NET, Rust, network protocols and reversing Windows internals 😊



# github.com/Devolutions/multi-pwsh



Wiesbaden  
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- Side-by-side PowerShell installs (7.4, 7.5, 7.6 incl. previews/RCs)
- Per-version aliases (e.g. pwsh-7.4.11), pin or track latest
- Virtual environments for PowerShell module isolation
- Export / import virtual environments as zip archives
- .NET native Rust host – not the regular .NET apphost.exe



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# Getting Started

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Bootstrap, Install, and Verify



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# Demo Time

Let's see it in action

[github.com/Devolutions/multi-pwsh](https://github.com/Devolutions/multi-pwsh)



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# Installing PowerShell with multi-pwsh

**Install and verify side-by-side versions:**

```
$ multi-pwsh install 7.4  
$ multi-pwsh install 7.5  
$ pwsh-7.4 --version | pwsh-7.5 --version
```

**Flexible version selectors:**

7 = latest 7.x | 7.4 = latest 7.4.x | 7.4.11 = exact | 7.4.x = all patches

PowerShell 7.6.1

PS C:\Users\mamoreau> multi-pwsh install 7.7 --include-prerelease

Installed PowerShell 7.7.0-preview.1

Scope: user

Install root: C:\Users\mamoreau\AppData\Local\PowerShell

Version path: C:\Users\mamoreau\AppData\Local\PowerShell\7.7.0-preview.1

Updated patch alias: C:\Users\mamoreau\AppData\Local\PowerShell\bin\pwsh-7.7.0-preview.1.exe

Updated alias: C:\Users\mamoreau\AppData\Local\PowerShell\bin\pwsh-7.7.exe

Updated major alias: C:\Users\mamoreau\AppData\Local\PowerShell\bin\pwsh-7.exe

Alias bin: C:\Users\mamoreau\AppData\Local\PowerShell\bin

Add to PATH once for this scope: C:\Users\mamoreau\AppData\Local\PowerShell\bin

PS C:\Users\mamoreau> pwsh-7.7 --version

PowerShell 7.7.0-preview.1

PS C:\Users\mamoreau> pwsh-7.7

PowerShell 7.7.0-preview.1

PS C:\Users\mamoreau> \$PSVersionTable

| Name                      | Value                        |
|---------------------------|------------------------------|
| PSVersion                 | 7.7.0-preview.1              |
| PSEdition                 | Core                         |
| GitCommitId               | 7.7.0-preview.1              |
| OS                        | Microsoft Windows 10.0.26100 |
| Platform                  | Win32NT                      |
| PSCompatibleVersions      | {1.0, 2.0, 3.0, 4.0...}      |
| PSRemotingProtocolVersion | 2.4                          |
| SerializationVersion      | 1.1.0.1                      |
| WSManStackVersion         | 3.0                          |

PS C:\Users\mamoreau>

# Scoped Installs & Alias Management

## User vs. Machine scope (Windows, macOS, Linux):

```
$ multi-pwsh install 7.5 --scope machine
```

Versions stay side-by-side under the selected install root

PATH only needs one entry per scope; aliases live in a stable bin dir

## Alias pinning and lifecycle:

```
$ multi-pwsh alias set 7.4 7.4.11    # pin to exact version
```

```
$ multi-pwsh alias unset 7.4        # unpin, track latest
```

## Windows integration flags:

```
--enable-psremoting --add-explorer-context-menu --register-manifest
```

# Virtual Environments & Native Host

## Isolated module roots (like Python venvs for PowerShell):

```
$ multi-pwsh venv create msgraph  
$ multi-pwsh host 7.4 -venv msgraph -NoLogo -NoProfile  
Save-Module directly into venv root; PSModulePath is scoped per launch
```

## Export and import for reproducible environments:

```
$ multi-pwsh venv export msgraph msgraph.zip  
$ multi-pwsh venv import msgraph-copy msgraph.zip
```

## Native host mode (pwsh-host crate):

Runs PowerShell through native hosting, not a subprocess

Alias shims are hard links to multi-pwsh; auto-repaired by doctor

# Q&A

15 minutes



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